

System programming

Assessment Test –1

Date: 15-6-26

Marks:50 M

Theory

(5*2=10M)

1. Explain the difference between a Process and a Thread. When would you choose threads over processes?
2. Explain how Virtual Address Translation works. Describe the roles of Virtual Memory, MMU, Page Tables, and TLB.
3. What is the different IPC (Inter-Process Communication) mechanisms available in Linux? Compare Pipe, FIFO, Message Queue, Shared Memory, Signals, and Sockets.
4. What is a Race Condition? How do Mutexes and Semaphores help in synchronization? Explain the difference between them.
5. Compare TCP and UDP communication. Explain the TCP Three-Way Handshake when UDP is preferred over TCP.

Coding Questions

(4*5 =20M)

1. Write a C program that creates a child process using fork(). The child should print numbers from 1 to 5, and the parent should wait for the child to complete before printing a message.
2. Write a program to handle the SIGINT signal (Ctrl+C). Instead of terminating immediately, the program should print a custom message whenever the signal is received.
3. Write a parent-child communication program using pipe(). The parent should send a string message to the child, and the child should display the received message.
4. Write a multi-threaded program where two threads increment a shared counter. First demonstrate the race condition, then fix it using a mutex.